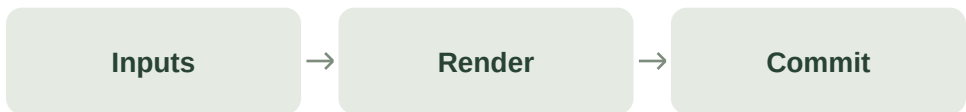


React Notes

A clearer model for React

A component describes the interface for a particular set of props, state and context. Rendering calculates that description; committing applies the necessary changes to the screen.



Keep rendering pure: do not change unrelated objects, start requests or subscribe to events while calculating the interface.

Reading question: what belongs in rendering, and what belongs in an event handler?

State is a snapshot

Setting state requests another render. It does not change the state variable inside the event handler that is already running.

```
setCount(count + 1);  
console.log(count);  
// Still the current render's value.
```

When the next value depends on the previous queued value, use a functional updater.

```
setCount(previous => previous + 1);  
setCount(previous => previous + 1);
```

The updater receives the previous pending value, not a magically mutated local variable.

Keys describe identity

A stable key helps React match an item to its previous instance within a list of siblings. Reordering should not make one item inherit the local state of another.

```
items.map(item => (  
  <Row key={item.id} item={item} />  
));
```

An array index can be a poor key when items move, are inserted or removed. A freshly generated random key also destroys continuity between renders.

Practice: explain why editing a row can reveal a bad key choice.

Return to the source

These are original, short sample notes, included so you can try Shelf without importing anything. They are not a complete React course.

Use a bookmark to keep this page. Select a useful sentence and highlight it, or add a note describing it in your own words.

Primary reading: react.dev/learn/render-and-commit
react.dev/learn/state-as-a-snapshot
react.dev/learn/queueing-a-series-of-state-updates
react.dev/learn/rendering-lists

Your highlights belong to you. Shelf keeps them separate from the original PDF.